

Bellabeat Wellness Case Study

Executive Summary

This case study analyzes Fitbit smart-device usage data to identify patterns in physical activity, sedentary behavior, and daily activity consistency and translate those findings into recommendations for Bellabeat's marketing strategy. Analysis of 35 users revealed substantial differences in activity behavior, including a large lower-activity segment, higher levels of sedentary time, greater day-to-day activity variability, and a much higher frequency of very-low-activity days among lower-activity users. Sustained activity was most commonly observed during the afternoon. Based on these findings, the analysis recommends that Bellabeat focus on reducing prolonged sedentary behavior, encouraging consistent daily movement, addressing very-low-activity days, and strategically timing user engagement around observed activity patterns.

Business Task & Objectives

The primary objective of this analysis is to identify patterns in smart-device usage and activity behavior and determine how these insights could inform Bellabeat's marketing strategy.

Key Business Questions

- What trends can be identified in smart-device usage and physical activity?
- How do these trends reveal differences in user behavior?
- How could these insights inform Bellabeat's marketing strategy?

Data & Sources

The primary dataset used for this case study is the Fitbit Fitness Tracker Data available through Mobius on Kaggle. The dataset contains activity and wellness observations from 35 users collected across two collection periods spanning March 12 through May 12, 2016. Available data includes daily, hourly, and minute-level measures of physical activity, intensity, steps, calories, sleep, heart rate, and weight-related measurements.

For this analysis, the Daily Activity, Hourly Intensities, Sleep Day, and Weight Log datasets were selected based on their relevance to the business questions and compatibility with the analysis. The data was prepared and validated before analysis using Python, Google Sheets, and SQL/BigQuery, with Tableau used to visualize key findings.

Data Limitations & Assumptions

The dataset has several limitations that should be considered when interpreting the findings:

- **Sample size:** The Daily Activity dataset contains 35 unique users. While sufficient for identifying patterns within the sample, the small sample limits how confidently the findings can be generalized to the broader population of smart-device users.
- **Sleep data availability:** Sleep Day data was available for only 24 of the 35 users and covered only the second collection period. Sleep-related findings therefore represent a smaller subset of participants.
- **Limited user context:** The dataset does not provide sufficient demographic or contextual information to determine whether differences in activity behavior are associated with factors such as age, occupation, or other individual characteristics.
- **Directional insights:** The analysis assumes that patterns observed within the sample can provide useful directional insights for Bellabeat, while recognizing that the sample may not be representative of Bellabeat's broader customer base.

Data Cleaning and Preparation

The source dataset contained more than 20 CSV files across two collection periods and included daily, hourly, and minute-level records. Python and pandas were used to inventory the files and assess their structure before selecting datasets relevant to the business questions.

The Daily Activity, Hourly Intensities, Sleep Day, and Weight Log datasets were combined across the collection periods where appropriate and prepared for analysis. Data cleaning and validation were performed using Google Sheets and included duplicate checks, resolution of overlapping records between collection periods, missing-value assessments, data-type verification, and validity checks for numeric and time-based measures.

Key cleaning steps included:

- **Duplicate removal:** Identified and removed redundant records from the Sleep Day and Hourly Intensities datasets.
- **Overlapping records:** Resolved duplicate observations caused by the overlap between collection periods, retaining the later-period records to prevent double-counting.
- **Missing values:** No missing values were identified in the Daily Activity, Sleep Day, or Hourly Intensities datasets. Missing Fat measurements in the Weight Log dataset were retained because the values were unavailable in the source data.
- **Data validation:** Checked numeric fields for negative or clearly invalid values and verified activity-duration and sleep-duration relationships for consistency.
- **Final activity dataset:** After removing overlapping Daily Activity records, the dataset contained 1,373 observations across 35 unique users.

Analysis & Methodology

Following data cleaning and validation, the analysis was conducted at both the daily-observation and user levels. Daily-level analysis was used to identify overall activity patterns, while user-level aggregation was used to compare behavioral differences across participants.

SQL and Google Sheets were used to aggregate activity measures, calculate descriptive statistics, segment users by activity level, and examine relationships between physical activity, sedentary behavior, consistency, and very-low-activity days. Correlation analysis was used to evaluate relationships between daily steps, calorie expenditure, and sedentary behavior. Users were segmented into lower- and higher-activity groups based on average daily steps, with users averaging fewer than 5,000 steps per day classified as lower-activity. These groups were then compared to identify differences in activity composition and behavioral consistency.

Tableau was used to create visualizations that communicate the primary findings and support the development of business recommendations for Bellabeat.

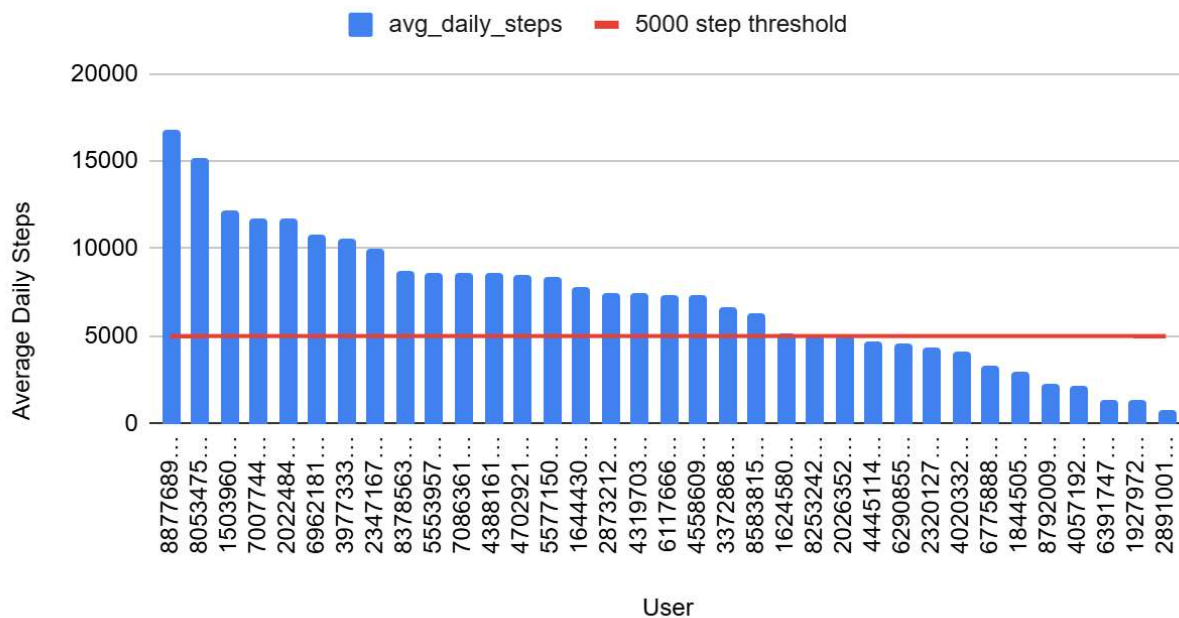
Key Findings & Visualizations

Finding 1: Activity levels vary substantially across users.

Key Finding: Activity levels varied substantially across the 35 users analyzed. Average daily steps ranged from 774 to 16,759 steps per day, and 11 users (31.4%) averaged fewer than

5,000 steps per day.

Average Daily Steps Varied Substantially Across Users



Interpretation: The wide range in average daily steps indicates substantial differences in activity behavior across users. Approximately one-third of the sample fell below 5,000 average daily steps, providing a meaningful lower-activity segment for further analysis.

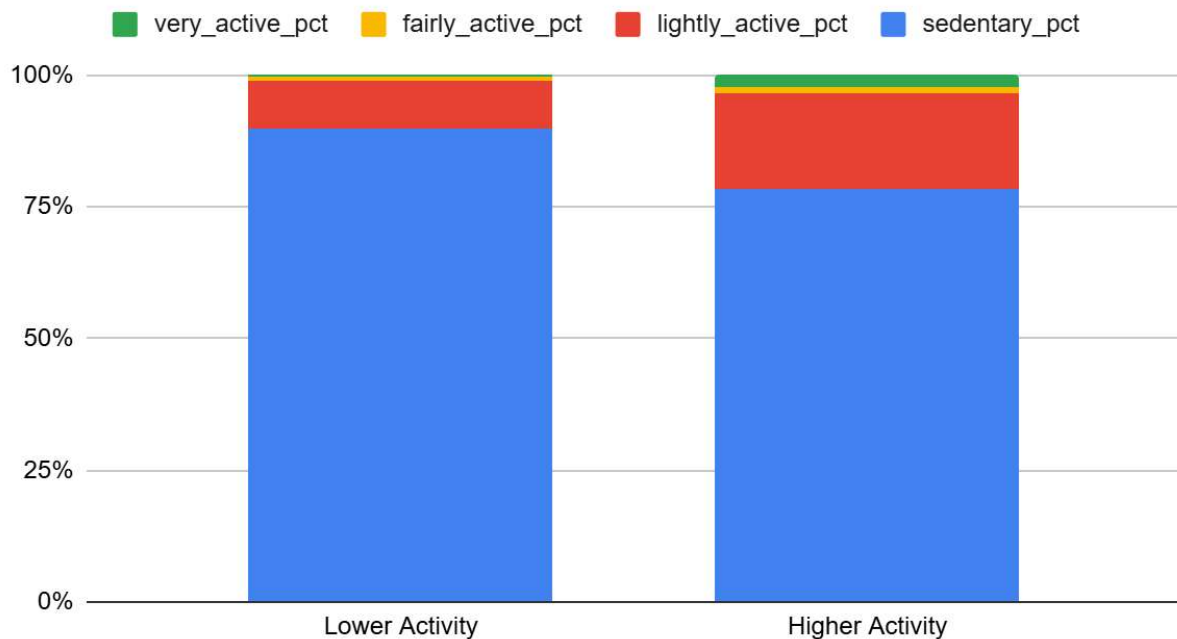
Business Implication: The variation in activity levels suggests that Bellabeat should avoid relying on a single activity target for all users. Personalized goals and engagement strategies could better support users at different activity levels.

Finding 2: Daily activity composition differs between lower- and higher-activity users.

Key Finding: Lower-activity users spent an average of 81.3% of the day sedentary, compared with 65.2% among higher-activity users. Higher-activity users also spent a greater proportion of the day lightly active (15.0%), fairly active (1.1%), and very active (1.8%), compared with 8.1%,

0.8%, and 0.2%, respectively, among lower-activity users.

Daily Activity Composition by Activity Level



Interpretation: The difference between lower- and higher-activity users extended beyond periods of intense exercise. Higher-activity users spent substantially less time sedentary and more time engaged in movement across every recorded activity category. This suggests that differences in overall activity may be influenced by everyday movement, not only dedicated exercise.

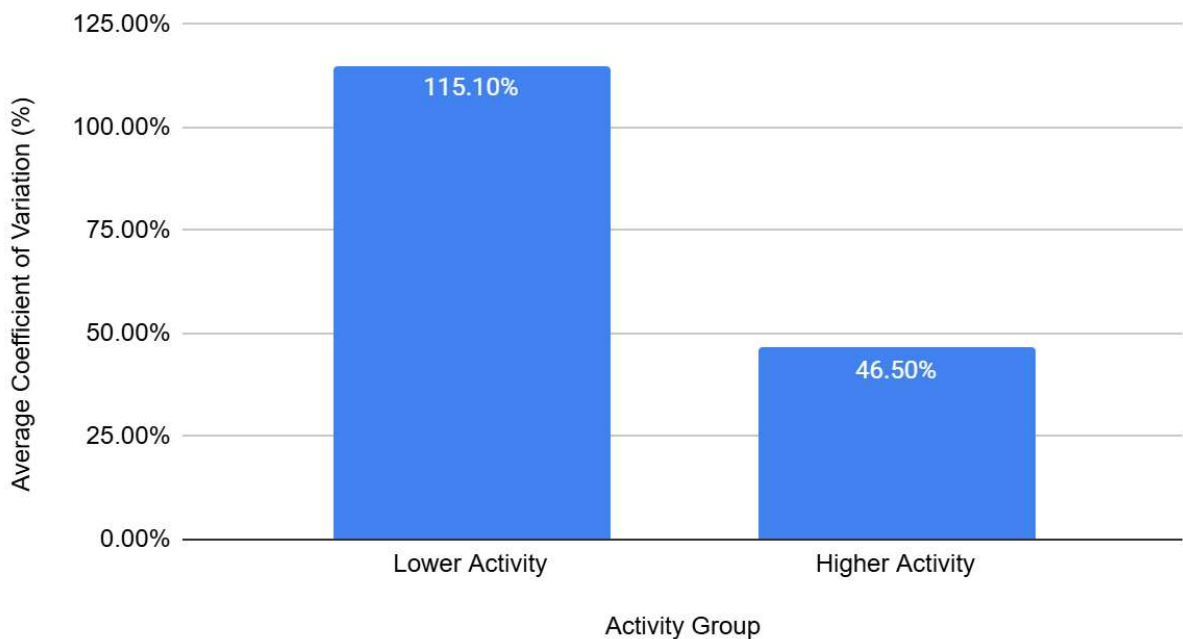
Business Implication: Bellabeat could encourage movement throughout the day rather than focusing exclusively on intense exercise or step targets. Personalized reminders to stand, walk, stretch, or take brief movement breaks could provide achievable opportunities for lower-activity users to increase daily movement and reduce sedentary time.

Finding 3: Higher-activity users maintain more consistent daily activity.

Key Finding: Lower-activity users had an average coefficient of variation (CV) of 115.1% in daily steps, compared with 46.5% among higher-activity users. The median CV showed a

similar pattern, at 110.0% for lower-activity users and 44.2% for higher-activity users.

Higher-Activity Users Had More Consistent Daily Activity



Interpretation: Higher-activity users demonstrated substantially less day-to-day variation in their activity levels relative to their typical daily activity. This suggests that higher activity in this sample was associated not only with taking more steps, but also with maintaining a more consistent daily movement pattern.

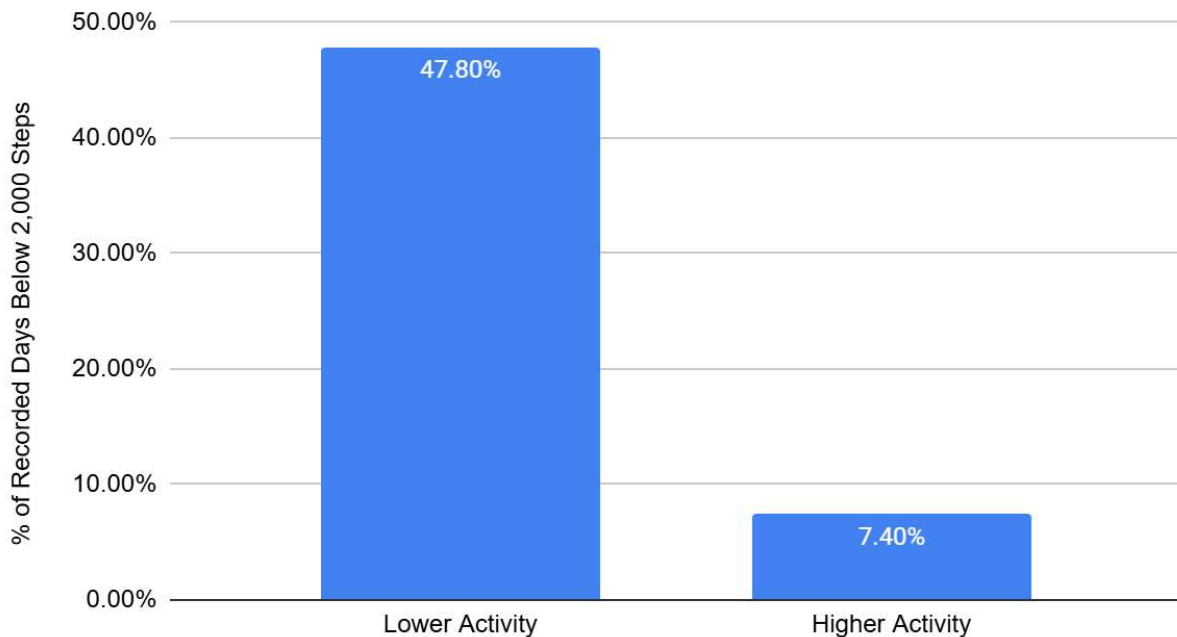
Business Implication: Bellabeat could encourage consistency by emphasizing achievable weekly movement goals rather than focusing exclusively on a single daily step target. Personalized goals that reward users for maintaining activity across multiple days could help support more sustainable movement habits.

Finding 4: Lower-activity users experience very-low-activity days more frequently.

Key Finding: Lower-activity users averaged 47.8% of their recorded days below 2,000 steps, compared with only 7.4% among higher-activity users. The median comparison showed a similar pattern, with 54.5% of recorded days below 2,000 steps for lower-activity users

compared with 4.9% for higher-activity users.

Lower-Activity Users Had More Very Low-Activity Days



Interpretation: Lower activity in this sample was not limited to having a lower average step count. Lower-activity users also experienced very-low-activity days far more frequently than higher-activity users. This suggests that inconsistent periods of extremely low movement may be an important behavioral characteristic of the lower-activity segment.

Business Implication: Bellabeat could identify unusually inactive days and provide personalized prompts to encourage movement. Rather than focusing exclusively on ambitious step targets, the app could help users establish a minimum level of daily movement and gradually increase that baseline over time.

Supporting Insight: Afternoon Activity Patterns

Among the 34 users with sufficient hourly activity data, 22 (64.7%) demonstrated their highest sustained activity during the afternoon. This pattern suggests that afternoon engagement may be a useful opportunity for Bellabeat to deliver activity prompts or challenges.

Recommendations/Action Plan

1. Reduce Sedentary Time

Use personalized inactivity reminders to encourage users to stand, walk, stretch, or take brief movement breaks throughout the day.

2. Address Very-Low-Activity Days

Identify unusually inactive days and provide personalized prompts focused on achieving a minimum level of daily movement rather than an ambitious step target.

3. Encourage Consistent Daily Movement

Introduce personalized weekly movement goals that reward users for maintaining activity across multiple days and building sustainable habits.

4. Leverage Afternoon Activity Patterns

Use observed afternoon activity patterns to strategically time movement reminders, challenges, and other engagement opportunities.

Action Plan

Bellabeat could incorporate these strategies into its app through personalized inactivity reminders, minimum-movement goals, weekly activity targets, and strategically timed afternoon engagement. The effectiveness of these interventions should be evaluated through user engagement and changes in activity behavior over time. Because this analysis is based on a relatively small sample of Fitbit users, these recommendations should be treated as opportunities for testing rather than definitive conclusions about all Bellabeat customers.

Conclusion

This analysis identified meaningful differences in activity behavior across the Fitbit users studied. Lower-activity users were more sedentary, demonstrated greater day-to-day variability, and experienced very-low-activity days far more frequently than higher-activity users. These patterns suggest that opportunities to increase everyday movement may extend beyond traditional exercise goals.

For Bellabeat, the findings support a strategy centered on reducing sedentary behavior, encouraging consistent movement, addressing unusually inactive days, and using observed activity patterns to guide the timing of user engagement. While the dataset is limited in size and

does not represent Bellabeat customers directly, the findings provide useful directional insights that Bellabeat could use to develop and test targeted engagement strategies.